Auteur: Anna Voronovska
Studierichting: Informatica
Oefeningenreeks 5A nr. 11.10

Bereken de volgende bepaalde integralen door partiële integratie

$$\int_0^{\frac{\pi}{2}} x^2 \sin x \cos^3 x \cdot dx$$

onbepaalde integraal:

$$\begin{cases} u = x^2 \\ dv = \sin x \cos^3 x \cdot dx \end{cases} \Rightarrow \begin{cases} du = 2x \cdot dx \\ v = -\frac{1}{4}\cos^4 x \end{cases}$$

$$I = -\frac{1}{4}x^2 \cos^4 x + \frac{1}{2}\int x \cos^4 x \cdot dx$$

Dubbel toepassen van de halveringsformule geeft:

$$\int x \cos^4 x \cdot dx = \int x \left(\frac{3}{8} + \frac{1}{2}\cos(2x) + \frac{1}{8}\cos(4x)\right) \cdot dx$$

$$= \frac{3}{16}x^2 + \frac{1}{2}\int x \cos(2x) \cdot dx + \frac{1}{8}\int x \cos(4x) \cdot dx$$

De termen $\int x \cos(ax) \cdot dx$:

$$\int x \cos(ax) \cdot dx = \frac{x}{a}\sin(ax) + \frac{1}{a^2}\cos(ax)$$

$$I = -\frac{1}{4}x^2 \cos^4 x + \frac{3}{32}x^2 + \frac{x}{8}\sin(2x) + \frac{1}{16}\cos(2x) + \frac{x}{64}\sin(4x) + \frac{1}{256}\cos(4x) + c$$

De bepaalde integraal:

$$I = \left[-\frac{1}{4}x^2 \cos^4 x + \frac{3}{32}x^2 + \frac{x}{8}\sin(2x) + \frac{1}{16}\cos(2x) + \frac{x}{64}\sin(4x) + \frac{1}{256}\cos(4x)\right]_0^{\frac{\pi}{2}}$$

bovengrens: $\frac{\pi}{2}$:

$$\left(0 + \frac{3\pi^2}{128} + 0 - \frac{1}{16} + 0 + \frac{1}{256}\right) = \frac{3\pi^2}{128} - \frac{15}{256}$$

ondergrens: 0:

$$\left(0 + 0 + 0 + \frac{1}{16} + 0 + \frac{1}{256}\right) = \frac{17}{256}$$

$$I = \left(\frac{3\pi^2}{128} - \frac{15}{256}\right) - \frac{17}{256} = \frac{3\pi^2}{128} - \frac{32}{256} = \frac{3\pi^2}{128} - \frac{1}{8}$$

References